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Generalized crack initiation and crack damage stress thresholds of brittle rock masses near underground excavations

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Abstract

The rock mass failure process is characterized by several distinct deformation stages which include crack initiation, crack propagation and coalescence. It is important to know the stress levels associated with these deformation stages for engineering design and practice.

Extensive theoretical, experimental and numerical studies on the failure process of intact rocks exist. It is generally understood that crack initiation starts at 0.3 to 0.5 times the peak uniaxial compressive stress. In confined conditions, the constant-deviatoric stress criterion was found to describe the crack initiation stress level.

Here, generalized crack initiation and crack damage thresholds of rock masses are proposed. The crack initiation threshold is defined by $\sigma_1 - \sigma_3 = A\ \sigma_{cm}$ and the crack damage threshold is defined by $\sigma_1 - \sigma_3 = B\ \sigma_{cm}$ for jointed rock masses, where A and B are material constants and σ_{cm} is the uniaxial compressive strength of the rock masses. For a massive rock mass without joints, σ_{cm} is equal to σ_{cd}, the long-term uniaxial strength of intact rock. After examining data from intact rocks and jointed rock masses, it was found that for massive to moderately jointed rock masses, the material constants A and B are in the range of 0.4 to 0.5, 0.8 to 0.9, respectively, and for moderately to highly jointed rock masses, A and B are in the range of 0.5 to 0.6, 0.9 to 1.0, respectively. The generalized crack initiation and crack damage thresholds, when combined with simple linear elastic stress analysis, assist in assessing the rock mass integrity in low confinement conditions, greatly reducing the effort needed to obtain the required material constants for engineering design of underground excavations.
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1. Introduction

The initiation and propagation of stress induced crack damage is a necessary precursor to all brittle failure of rocks, eventually leading to borehole breakout, notch formation, and abrupt failure of underground excavations in mining and civil tunneling. It is therefore important to have a suitable criterion to predict the onset of such damage for safe engineering of underground works.

Although stress induced damage can be directly observed using electron microscopes [1–3], direct visual observation has found limited application in practical engineering for underground construction when rock mass damage initiation needs to be observed. Hence, indirect monitoring methods, such as the acoustic emission (AE) and microseismic (MS) monitoring techniques, have been introduced in rock engineering for rock damage assessment both in the laboratory and in situ. AE event emissions are sound waves emitted by microcracks or fractures as they are created or propagated. The sound waves propagate through the sample or rock and are recorded by sonic sensors connected to an acquisition system. Processing techniques, similar to those used by seismologists, are applied in order to determine information concerning the AE rate, locations and failure mechanism. AE counts and magnitudes (energy release) show the amount and intensity of fracturing whereas AE locations delineate regions of damage. Microcrack distributions, mapped three-dimensionally through time, describe damage accumulation, crack coalescence and macro-fracture

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propagation. AE/MS techniques have been used by many, e.g., in mining for rockburst risk assessment [4], roof fall prediction [5], and in civil engineering for rock slope stability, rock damage evaluation, and large-scale underground cavern stability monitoring [6,7].

The failure of brittle rocks in the laboratory has been extensively studied (e.g. [8–12 and others]). The monitoring of AE waves in the laboratory, usually in the 200–2000 kHz range, has proven to be a powerful tools available to study the brittle failure of rocks in a controlled stress environment. There is generally a good correlation between AE rate and inelastic strain rate so that the AE rates can be used to quantify damage initiation and accumulation in rock sample (e.g. [13]).

Besides the laboratory study of AE in rock samples, there have been many works conducted in situ to study AE/MS. A very well planned and executed experiment is the Mine-by Experiment at the URL in Canada [14]. The main objective of the experiment was to study processes involved in progressive failure of brittle rock and the development of excavation-induced damage around underground openings. The details of this project are presented in an extensive report by Read and Martin [14].

Based on an analysis of the monitoring data, a number of phenomenological criteria or the so-called crack initiation or damage initiation thresholds have been proposed to predict the onset of rock damage [15–17]. It is observed that these criteria work well for intact and massive rock masses when damage initiation occurs at about 1/3 of uniaxial compressive strength. For example, the crack initiation stress level for the URL is $\sigma_1 - \sigma_3 = 0.33\ \sigma_c$ [15]. However, for jointed rock masses, these criteria fail to reach a consistent assessment as for example encountered at the ZEDEX tunnel, the crack initiation stress level is much lower at $\sigma_1 - \sigma_3 = 0.13\ \sigma_c$ [18].

In the present study, experimental and theoretical study of crack damage in intact rocks are reviewed and discussed. The crack initiation and crack damage thresholds proposed and used for intact rocks are then generalized for jointed rock masses. The deviatoric stress criterion is used to describe the crack initiation and crack damage surfaces in the principal stress space. The generalization is based on in situ rock mass damage assessment using AE/MS monitoring and direct visual observation, combined with detailed rock mass quality assessment techniques. Data from several well-documented case studies are analyzed to demonstrate the validity of this generalization. The terms "crack initiation" and "crack damage" are used to describe the onset of damage (initiation) and the onset of crack coalescence to form macroscopic fracture (coalescence), equivalent to "AE initiation" and "AE damage," respectively.

2. Microfracturing, damage and failure of brittle intact rocks

2.1. Experiment data

Rocks inevitably contain pores, flaws or microcracks. For example, weak grain boundaries may serve as microcrack source. These microcracks will propagate when the rock is loaded beyond a certain threshold and this is associated with acoustic emissions. Several characteristic stress levels have been identified from laboratory test and are important in understanding the damage process of brittle rocks. In the stress–strain relation shown in Fig. 1, σ_{cc} is the crack closure stress level, σ_{ci} is the crack initiation stress level, and σ_{cd} is the crack damage stress level, which corresponds to the long-term rock strength [15,19,20]. AE rates start at σ_{ci} and increase drastically when σ_{cd} is reached. The maximum AE rate is usually reached near the peak strength σ_c. σ_c depends on the loading rate; usually, the higher the loading rate, the higher σ_c. The three stress levels, i.e., σ_{ci}, σ_{cd}, and σ_c, represent important stages in the development of the macroscopic failure process of intact rocks.

The characteristics identified from laboratory test results are:

(1) Crack (AE) initiation starts at stress levels of approximately 0.3–0.5 times the peak uniaxial load [9,10,19,21].
(2) Microcracks form or propagate mainly in the direction parallel to the maximum compressive stress. This is evident from the volumetric strain dilation [12], direct microscopic observation of the test samples [3], and numerical (e.g., PFC) modeling results [22].
(3) Onset of fracture coalescence starts at stress levels of approximately 0.7–0.8 times the peak strength [11,12].
(4) Macrocracks or shear bands formation normally follows after the peak strength is reached.

In laboratory tests on intact rocks, the crack initiation stress or threshold is defined by the onset of stable crack growth. It is sometimes defined as the point where the crack volumetric strain deviates from zero. AE starts to appear systematically when the stress level is above σ_{ci}. In some cases, where the samples have experienced sampling damage, AE can be detected at the crack closure stage. When accumulated AE counts are used to study the crack initiation and crack damage stress levels, AE events recorded during the crack closure must be subtracted from the total cumulative count to avoid confusion in the data analysis [23].

Existing cracks start to propagate and new cracks initiate in a stable fashion when the damage initiation limit is exceeded but critical rock mass damage is only

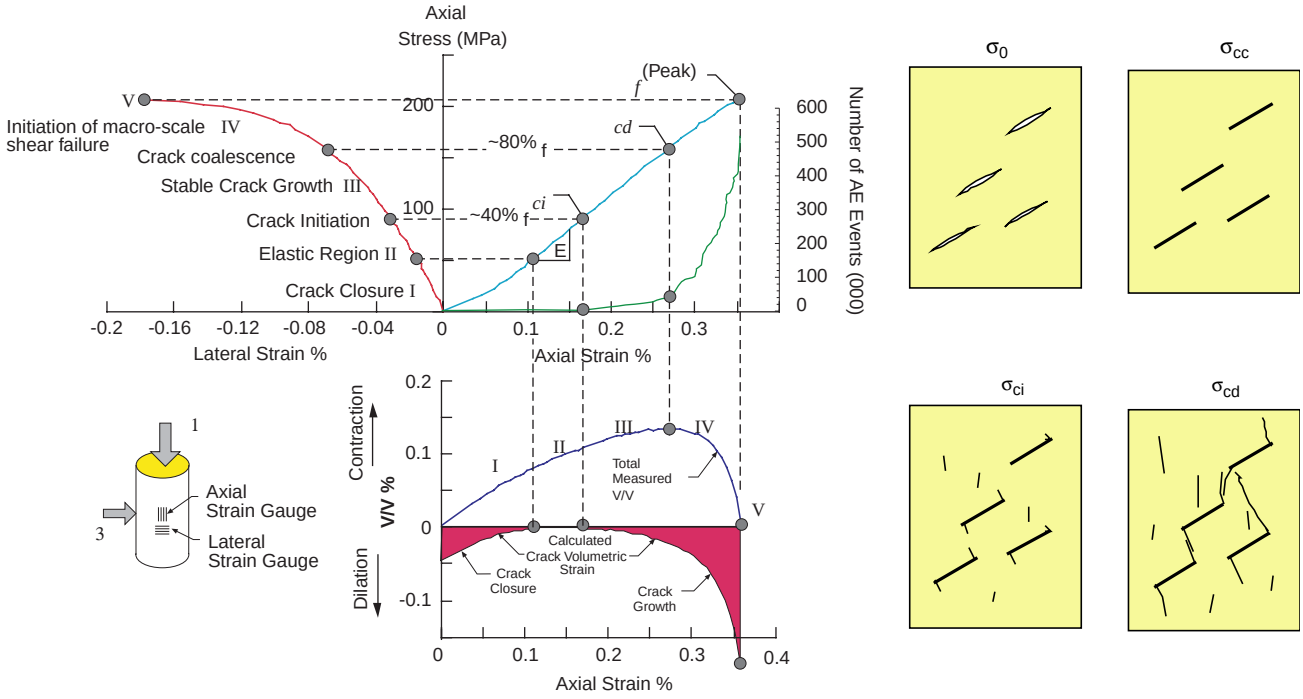


Fig. 1. Stress–strain diagram showing the stages of crack development (after Martin et al. [20]). A schematic representation of the different stress levels using microcrack initiation, propagation and coalescence is shown to the right.

Table 1
Summary of crack initiation and crack damage stress levels from laboratory tests

Rock type	σ_3 (MPa)	σ_{ci} (MPa)	σ_{cd} (MPa)	σ_c (MPa)	σ_{ci}/σ_c	σ_{cd}/σ_c	σ_{ci}/σ_{cd}	References
Sandstone	2	34	59	70	0.49	0.84	0.58	[24]
Granite	0	81.5	156	206.9	0.39	0.75	0.52	[23]
Berea sandstone	7.5	23	—	44	0.50			[25]
Granite	41	245	515	613	0.40	0.84		[26]
Dolomites	0	165	274	274	0.60	1.0	0.60	[27]
Dolomites	10	90	110	154	0.58	0.71	0.82	[27]
Quartzite	0	114	241	283	0.40	0.85	0.47	[10]
Granite	0	80	180	224	0.36	0.80	0.50	[19]
Sandston	0	121	170	234	0.52	0.73	0.71	[9]
Dolerite	0	140		230	0.60			[2]
Granite	4.9	90	160	225	0.40	0.71	0.56	[28]
Granite	60	390		720	0.54			[29]

encountered when the crack density is sufficient for cracks to coalesce to form shear band or tensile spalls. This state is defined as the "crack damage stress" [12] or "AE damage threshold", because in general, a single AE event can be viewed as a cracking event. The crack damage stress is generally defined as the point where volumetric strain reversal occurs and unstable crack growth begins. It also corresponds to the stress where a drastic increase of AE activity (the slope of the accumulated AE events increase sharply) is observed. In laboratory test, the crack damage stress level σ_{cd} increases as confinement increases but in underground excavation, the tangential stress at the tunnel wall where there is little or no confinement drives fracture. Hence, the unconfined σ_{cd} determines the onset of failure propagation process in the field. In other word, σ_{cd} can be considered as the "wall rock strength" in massive rocks.

Table 1 summarizes some of the crack initiation and crack damage stress levels obtained from uniaxial and triaxial compression tests. In triaxial tests, the stress levels are defined by the deviatoric stress ($\sigma_1-\sigma_3$). It is seen that the σ_{ci}/σ_c ratio varies between 0.36 and 0.60 and the σ_{cd}/σ_c ratio varies between 0.71 and 1.0.

In uniaxial compression, the crack initiation stress level for most rocks falls in the range of 0.3 to 0.5. For confined rock, the damage initiation criterion is often described by a constant-deviatoric stress limit. Holcomb

and Costin [21] proposed $\sigma_1 - \sigma_{3=}0.33\ \sigma_c$ for brittle rocks. For Lac du Bonnet granite, the crack initiation stress level is defined by Martin [19] as

$$\sigma_1 - \sigma_3 = (0.4 \pm 0.05)\sigma_c. \quad (1)$$

In other words, the material behave at the damage initiation stress level like a frictionless material.

Other study results indicate that the crack initiation stress level of intact rocks can be approximated by [30–32]

$$\sigma_1 - (1.5\ \text{to}\ 2)\sigma_3 = 0.4\sigma_c. \quad (2)$$

The major difference between Eq. (2) and Eq. (1) is that the surface represented by the later has a slightly steeper slope in the principal stress space, suggesting that some frictional strength is mobilized at higher initiation stress level in brittle rocks. Near the underground excavation boundary, the confining stress is low so that the slope of the crack initiation surface is of subordinate importance. In the immediate vicinity of an excavation wall, the confining stress varies between 0 and a few MPa. In this range, the effect of the slope only adds a small percentage to the threshold level.

In summary, the crack (AE) initiation threshold for intact rocks in low confinement area can be written in the form of

$$\sigma_1 - \sigma_3 = A\sigma_{cd}. \quad (3)$$

Eq. (3) defines the crack initiation threshold by two material constants A and σ_{cd}. Because σ_c is not a material constant, depending for example on the loading rate, it is meaningful to normalize to the long-term strength described by the crack damage stress σ_{cd}. The ratio of σ_{ci}/σ_{cd} is equivalent to the constant A. In most cases, A is in the range of 0.4–0.6 (Table 1). As discussed in the next section that the micromechanics based models, such as those proposed by Kachanov [33], Horii and Nemat-Nasser [34] and Ashby and Hallam [35] systematically lead to such linear functions in the $\sigma_1 - \sigma_3$ space.

2.2. Theoretical model interpretation of crack initiation threshold

Rocks usually contain randomly oriented grains and flaws in the form of microcracks. Cracks start to propagate when a fracture criterion is met. The crack initiation mechanism can be studied using three major criteria for solids containing microcracks, i.e., stress-based criteria (e.g. the maximum tangential stress criterion by Erdogan and Sih [36]), energy-based criteria (e.g. the minimum strain energy density criteria by Sih [37]) and strain-based criteria (e.g. the extension strain criterion by Stacey [38]).

The first criterion for crack initiation was proposed by Griffith [39], who states that fractures occur when the energy stored in the rock overcomes the surface energy of the rock material. This criterion considers mode I cracking only and assumes the direction of crack initiation. Since its publication, the Griffith criterion had been extended to mode II fractures and modified to accommodate more complex loading conditions.

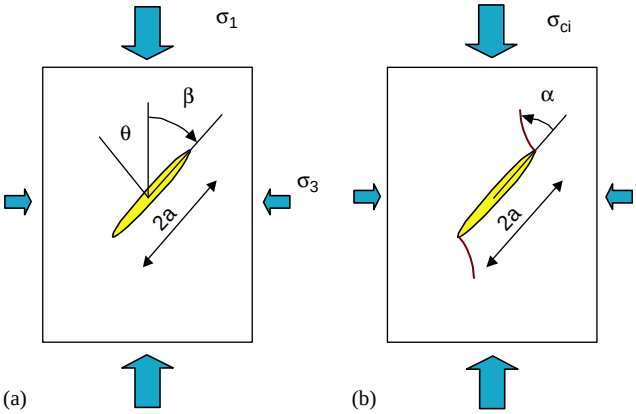


Fig. 2. Geometry of the (a) initial crack and (b) wing-crack initiation under compression.

Sih [37,40] used a strain energy density factor to successfully predict the crack initiation and direction. The theory is based two fundamental hypotheses, i.e., the critical intensity S_{cr} of the strain energy governs the onset of crack propagation and the crack initiation will start in a radial direction along which the strain energy density is a minimum. This theory is able to predict the wing-crack angle satisfactorily and for rocks with a Poisson's ratio of 0.25, it predicts a critical crack angle of $\beta = 37^\circ$ (Fig. 2). The crack initiation stress level is approximately equal to $\sigma_{ci} \approx \sqrt{59GS_{min}/a}$, where G is the shear modulus, a is the half-length of the initial crack and S_{min} is the minimum strain energy which is a material constant.

Cracks oriented between $\beta = 30{-}45^\circ$ are favorably oriented for crack initiation, according to Sih's theory. If S_{min} was linked to the rock tensile or uniaxial compressive strength, then the crack initiation stress level in terms of σ_{ci}/σ_c could be established. Unfortunately, this linkage is not available to the authors' awareness and the sliding crack model must be adopted to interpret the crack (AE) initiation stress level in brittle rocks.

The sliding crack model, for which an analytical solution is available, has been used widely to explain the brittle failure process of ceramics and rocks [34,35,37,41,42]. Prior to the initiation of the wing crack, the stress intensity factor at the tip of the crack is

$$K_{II} = \tau\sqrt{\pi a}, \quad (4)$$

where

$$\tau = \tfrac{1}{2}[(\sigma_1 - \sigma_3)\sin 2\theta - \mu(\sigma_1 + \sigma_3 + (\sigma_1 - \sigma_3)\cos 2\theta)], \quad (5)$$

where μ is the friction coefficient, σ_1 is the vertical compressive stress and σ_3 is horizontal confining stress. When the driving shear stress τ exceeds a critical value, two kinked cracks start to propagate. For a short nascent wing crack under compression, the stress intensity factor is [41]

$$K_I = \frac{2}{\sqrt{3}} K_{II} = \frac{2}{\sqrt{3}} \tau \sqrt{\pi a}. \quad (6)$$

Crack initiation happens when $K_I = K_{IC}$, where K_{IC} is the fracture toughness of the rock. Here, K_{IC} can be related to the rock's tensile strength in the form of $\sigma_t = 6.88\ K_{IC}$ [43]. According to Griffith theory, the rock compressive strength is equal to 8 times of the tensile strength. The original Griffith theory had been extended by Murrell [44] to include the triaxial state of stress. Murrell criterion predicts that the rock compressive strength equals 12 times the tensile strength. From laboratory test data, it is found that the compressive to tensile strength ratio for most rocks range from 8 to 15. Very high σ_c/σ_t strength ratios (>15) are reported by Lama and Vutukuri [45] and Sheorey [46]. This is largely due to the fact that some tests are performed on samples with weak structures that are favorably oriented for compression but unfavorably oriented for tension. In this paper, we focus on crack initiation on the grain size scale where very high compressive to tensile strength ratios are not applicable. It is therefore reasonable to assume that $\sigma_c = (55\text{–}82)\ K_{IC}$ with $\sigma_c/\sigma_t = 8\text{–}12$, incorporating the correlation between σ_t and K_{IC} [43]. In the case of uniaxial compression, the crack initiation stress can therefore be defined as

$$\sigma_{ci} \approx \frac{\sqrt{3}}{(55\ \text{to}\ 82)\sqrt{\pi a}(\sin 2\theta - \mu(1+\cos 2\theta))} \sigma_c. \quad (7)$$

When the crack surface friction is ignored, the normalized crack initiation stress is a function of the initial half crack length a (which is related to grain size) and the rock compressive to tensile strength ratio (Fig. 3). This figure illustrates that for coarse grain rock samples ($a > 1$ mm), the normalized crack initiation stress level is between $A = 0.3$ and 0.55 and for rocks with medium grain size ($a \sim 1$ mm), the normalized crack initiation stress level between $A = 0.4$ and 0.6. For very fine grained rocks ($a \ll 1$ mm), A could be exceeding 0.6.

If the initial microcracks contains kinks, it is the kink crack propagation that generates AE. For a kink crack with an initial length of l, the stress intensity factor [42] is

$$K_I = \frac{2a\tau\cos\theta}{\sqrt{\pi l}} - \sigma_3\sqrt{\pi l}. \quad (8)$$

Crack initiation happens again when $K_I = K_{IC}$. Assuming that the initial wing crack length is $a/2$ and the effect of friction and confinement are ignored, the crack initiation stress σ_{ci}/σ_c level for the kink crack to propagate is between 0.41 and 0.58 for $a = 1\text{–}2$ mm.

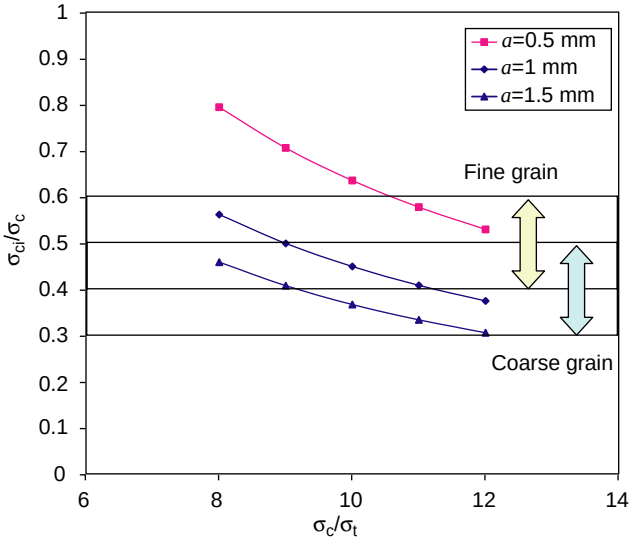


Fig. 3. Crack initiation stress level as a function of a and σ_c/σ_t ratio.

It follows that micromechanics based theories are capable of predicting the crack initiation stress level. The material constant A determined from these models is in good agreement with experimental results. The prediction of crack coalescence by these types of models is more difficult and the determination of σ_{cd} is therefore largely based on test results or empirical damage models calibrated on test data.

3. Damage initiation, propagation and failure of rock masses

3.1. Generalization of the crack initiation and crack damage thresholds

Rock masses are characterized by the existence of distributed joints and fractures, and the mechanical properties of jointed rock masses are thus strongly dependent on the properties and geometry of these weakness planes [47]. For jointed rock masses, there is no analytical solution available for the damage initiation, propagation, and coalescence thresholds. The determination of these threshold values is therefore largely based on direct monitoring and measurement such as AE/MS monitoring or constitutive models obtained from back-analysis of deformation monitoring data.

In moderately jointed rock masses, AE/MS monitoring results indicate that the damage process begins inside the rock masses, e.g., by breaking rock bridges between the existing joints and crushing of asperities. At this stress level, moderately jointed rock masses do not loose

their load carrying capacity and the deformation response is not sensitive enough to delineate the beginning of the damage process. In rock engineering practice, AE/MS monitoring has become the most effective method to determine the crack initiation threshold in situ.

In tunnels, cracking initiates as a result of the removal of the confining stress and the loading by tangential stress concentration. The cracks then tend to grow parallel to the walls forming slabs of rock. If there are no joints in the rock mass, such as in the massive granite at the URL, the stress-induced damage phenomena occur when the rock wall is loaded beyond its in situ unconfined compressive strength. The wall rock strength thus determines when visible failure (e.g., notch) results. By the same reasoning, the above argument can be expanded to jointed rock masses, i.e., the wall rock strength of the jointed rock mass controls the onset of rock mass damage. It is therefore important to characterize the uniaxial strength of a jointed rock mass for damage assessment.

In the rock mechanics community, the Hoek–Brown criterion is widely used to describe the peak strength of a jointed rock mass. The generalized Hoek–Brown criterion for jointed rock masses [48] is

$$\sigma_1 = \sigma_3 + \sigma_c\left(m_b\frac{\sigma_3}{\sigma_c} + s\right)^a, \tag{9}$$

where m_b, s, a are constants for the rock mass, and σ_c is the uniaxial compressive strength of the intact rock. For jointed rock masses, the constants can be obtained from rock mass classification using the geological strength index (GSI) system, i.e., [49]

$$m_b = m_i \exp\left(\frac{GSI - 100}{28 - 14D}\right), \tag{10}$$

$$s = \exp\left(\frac{GSI - 100}{9 - 3D}\right), \tag{11}$$

$$a = 0.5 + \frac{1}{6}(e^{-GSI/15} - e^{-20/3}), \tag{12}$$

where D is a factor that depends on the degree of disturbance to which the rock mass has been subjected by blast damage and stress relaxation ($D=0$ for careful excavation with controlled blasting). GSI values can be estimated based on the geological description of the rock mass and this is well suited for rock mass characterization without direct access to the rock mass at tunnel face. The GSI system concentrates on the description of two factors, structure and block surface conditions. A means to quantify this approach by use of field measurement data was recently presented by Cai et al. [50].

Pelli et al. [51] found that the parameters obtained from Eqs. (10) and (11) (with $D=0$) did not predict the observed failure locations and extent near a tunnel in a cemented sand or siltstone. They found that lower m_b and higher s values were required to match predictions with observations. Further analyses of underground excavations in brittle rocks eventually lead to the development of brittle Hoek–Brown parameters ($m_b=0$, $s=0.11$) by Martin et al. [52] for massive to moderately fractured rock masses with tight interlock that fail by spalling or slabbing rather than by shear failure. The fundamental assumption in using the brittle parameters is that the damage process around the tunnel is dominated by cohesion loss associated with rock mass damage through fracturing. Although this concept is only valid at low confinement stress, it explains the field observations very well.

$m_b = 0$, $s = 0.11$ means that tensile cracking causes crack dilation after damage initiation, which prevents the coincidental mobilization of friction and cohesion. The frictional strength can only be mobilized when the rock mass is sufficiently damaged to become essentially cohesionless. Substituting $m_b = 0$, $s = 0.11$ into Eq. (9) and assuming $D=0$, we obtain the crack damage stress threshold as

$$\sigma_1 - \sigma_3 = \sqrt{\exp^{(GSI-100)/9}}\sigma_c. \tag{13}$$

If the above equation is rewritten as

$$\sigma_1 - \sigma_3 = B\sigma_{cm}, \tag{14}$$

where B is a constant and σ_{cm} is the rock mass or wall rock strength which can be obtained from the GSI system. For a rock mass with $GSI=65$–100, and $\sigma_c=100$–250 MPa, Fig. 4 shows that the B value in Eq. (14) varies between 0.8 and 1.0.

One important generalization of Eq. (3) is that by replacing σ_{cd} with σ_{cm}, the in situ rock mass strength, Eq. (3) becomes applicable to damage initiation of jointed rock masses. The generalized crack (AE) initiation threshold is therefore defined as

$$\sigma_1 - \sigma_3 = A\sigma_{cm}, \tag{15}$$

and the generalized crack (AE) damage threshold is defined in Eq. (14). For intact rocks without joints, σ_{cm} is equal to the long-term UCS of the rock (σ_{cd}). A and B are constants depending on the rock mass property and can be considered as material constants. In general, A is in the range of 0.4–0.6 and B in the range of 0.8–1.0, for both intact rocks and jointed rock masses. When joints or foliation planes are unfavorably oriented such that failure occurs along the weak plane, the crack (AE) initiation stress level gets closer to the crack (AE) damage stress level ($A > 0.6$). This is supported by laboratory and field AE monitoring data discussed in the following sections.

For the application of these generalized crack initiation and crack damage thresholds, rock mass

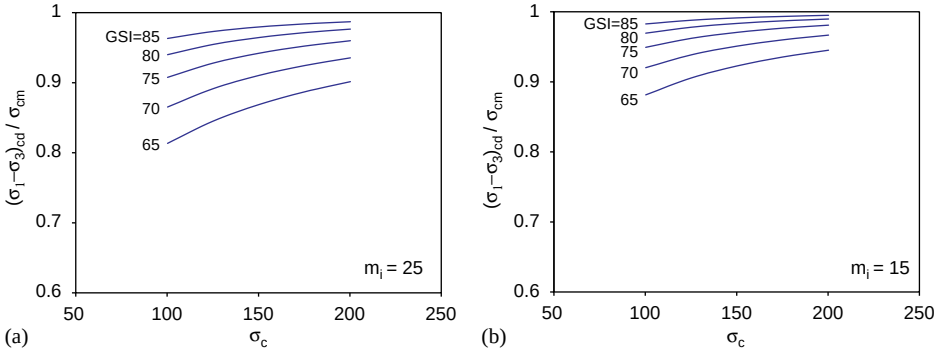


Fig. 4. Rock mass damage threshold level: (a) $m_i = 25$; (b) $m_i = 15$.

strength σ_{cm} can be obtained by the method presented by Hoek et al. [49]:

$$\sigma_{cm} = \frac{2c_m \cos\phi_m}{1 - \sin\phi_m}, \tag{16}$$

where c_m and ϕ_m are the equivalent cohesive strength and angle of friction of the rock mass, respectively.

This generalization of the crack initiation and crack damage thresholds has been verified through field monitoring results. Some examples are presented next.

3.2. Case histories

3.2.1. URL Mine-by tunnel

The URL Mine-by test tunnel, which was 46 m long and 3.5 m in diameter, was excavated roughly in the direction parallel to intermediate principal stress, using perimeter line drilling and mechanical breaking excavation method [14]. Extensive state-of-the-art geomechanical and geophysical instruments, including extensometers, convergence arrays, triaxial strain cells and AE/MS monitoring systems, were installed prior to the start of excavation to monitor the complete mechanical response of the rock mass surrounding the tunnel. An array of 16 triaxial accelerometers was installed to monitor excavation-induced seismicity near the advancing face. About 14% of the detected events (25,000) were source located. The spalling process produced v-shaped notch in the region of the maximum tangential stress concentration. It was observed that the damage process initiates about 0.3–0.4 diameters ahead of the tunnel face. 3D stress analysis revealed that when the differential stress level of $\sigma_1 - \sigma_3 = 70$–75 MPa was reached, the microseismic activity was observed [15].

The slabbing in the roof of the Mine-by tunnel was observed to start at about 1 m from the face, and failure progressed as the tunnel face was advanced [15]. Because the tangential stress was higher at the wall surface, the stress level at which the visible crushing started must represent the in situ rock wall strength. 3D elastic stress analyses were carried out to track the stress state at these critical points [53]. The stress path for a point located 10 mm from the tunnel wall at the maximum compressive stress concentration in the roof shows that the maximum tangential stress is about 120–147 MPa 1 m from the tunnel face, and eventually reaches 159 MPa when plane strain condition is reached far from the face. From vertical boreholes of different sizes the in situ rock mass strength was found to be between 150 and 160 MPa [53]. The reason why visible damage was observed at a lower computed stress level in the tunnel can be attributed to the irregularity of the surface of the Mine-by tunnel. Unlike the vertical boreholes excavated by driller raiser bores, the line drilling holes created local stress raisers which lead to premature failure. This is illustrated by the stress analysis results presented in Figs. 5 and 6. The maximum tangential stress is 169 MPa ($\sigma_1^0 = 60$ MPa, $\sigma_3^0 = 11$ MPa) for the tunnel with a smooth wall and 195 MPa at the local tangential stress concentrations (see the inset in Fig. 6). The local stress concentrations lead to a 15% higher stress, promoting early notch formation. Because the rock mass strength of 159 MPa is reached first at these local stress concentration points, the apparent smooth wall equivalent in situ rock mass strength is lower on the order of 120–147 MPa. If the Mine-by tunnel were driven by TBM, the notch formation would start further back from the face at about 3.5 m (one diameter of the tunnel) from the tunnel face.

Without loosing generality, the in situ rock or rock mass strength σ_{cm} of the granite at the URL at 159 MPa, or about 70% of laboratory strength ($\sigma_c = 224$ MPa) can be used. This leads us to a damage (AE/MS) initiation stress level of $\sigma_1 - \sigma_3 = 0.44$–$0.47\sigma_{cm}$ ($A = 0.44$–0.47). The upper limit, or the crack damage threshold, is identified from visual observations of notch formation. The notch formation is a gradual process. Only when the stress path crossed $\sigma_1 - \sigma_3 = 143$–$159$ MPa, or $\sigma_1 - \sigma_3 = 0.9$ to $1.0\sigma_{cm}$ ($B = 0.9$–1.0), did notch failure initiate and gradually propagate [15].

Martin et al. [52] suggested that the constant deviator criterion ($\sigma_1 - \sigma_3 = 70$ MPa, or $\sigma_1 - \sigma_3 = 0.3\sigma_c$) for the crack initiation provides an adequate approximation of the depth of the notch when applied to a complete plane

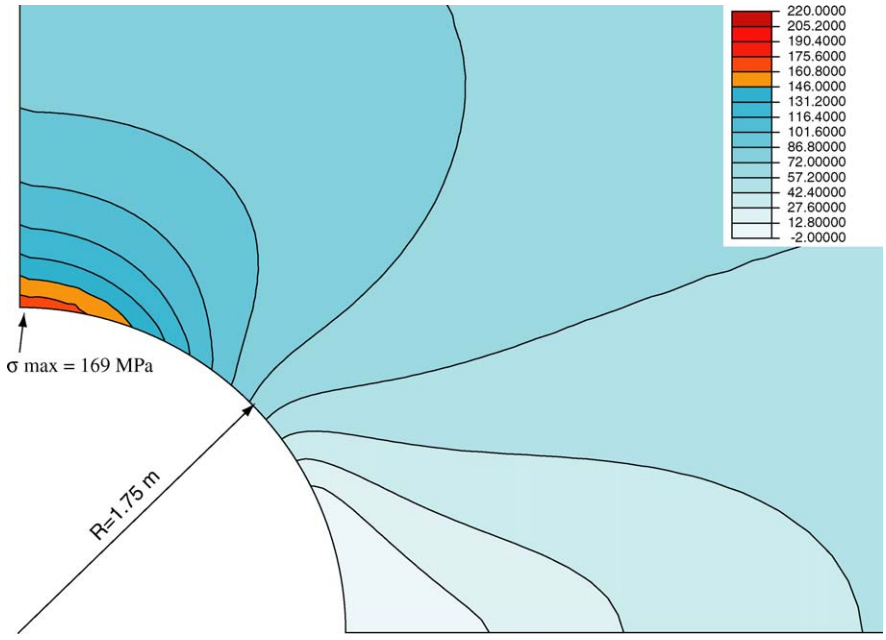


Fig. 5. σ_1 principal stress distribution around the Mine-by tunnel with smooth wall surface. Advantage is taken of the double symmetry in the tunnel configuration to limit the calculation time.

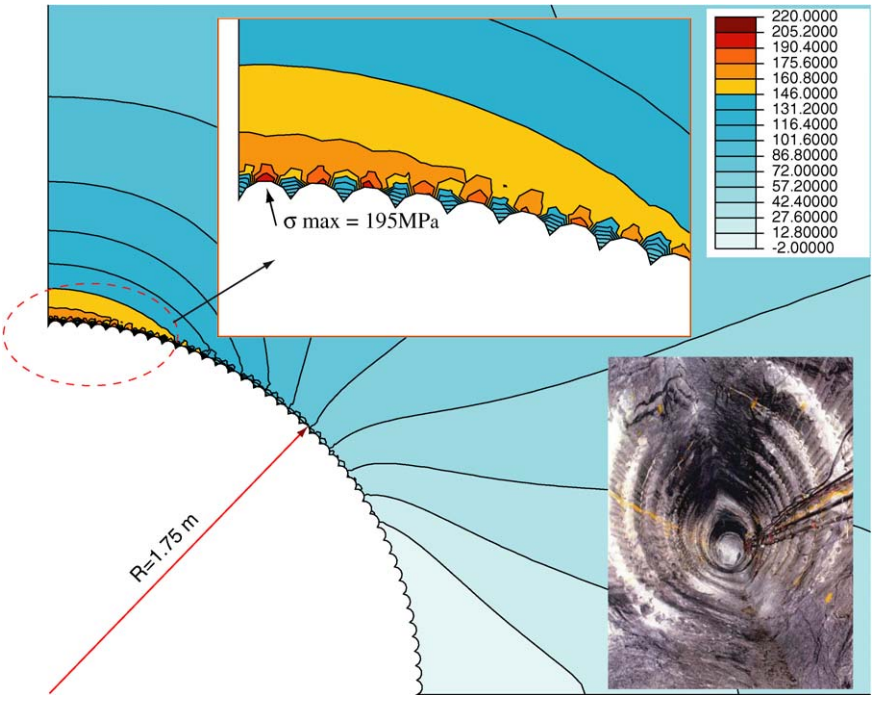


Fig. 6. σ_1 principal stress distribution around the Mine-by tunnel with non-smooth wall surface. The upper insert shows the detailed stress distribution and the lower insert shows the Mine-by tunnel with non-smooth wall surface.

strain condition. However, this threshold is not equal to the crack (AE) damage threshold.

Three crack (AE) damage thresholds are presented in Fig. 7. At URL, lab intact rock long-term strength σ_{cd} and in situ rock mass strength σ_{cm} are very close to each other (159 MPa), not by coincidence but rather because the local σ_{cd} represents the in situ strength of the massive rock mass. The failure surface of $\sigma_1-6\sigma_3=\sigma_{cm}$ is a good approximation of the average of all the lab intact rock long-term strength over a wide range of confine-

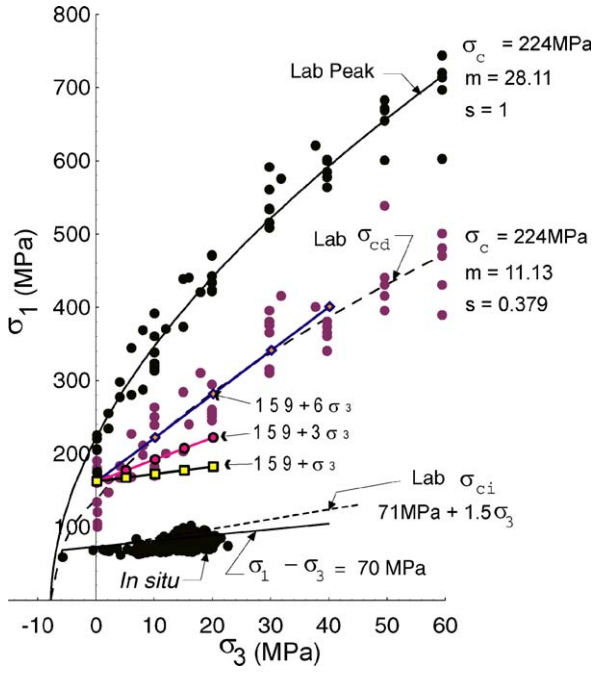


Fig. 7. Crack initiation stress threshold at the URL [15]. Crack damage thresholds of $\sigma_1 - \sigma_3 = 159$ MPa, $\sigma_1 - 3\sigma_3 = 159$ MPa, $\sigma_1 - 6\sigma_3 = 159$ MPa are added to the figure.

ment. $\sigma_1 - 3\sigma_3 = \sigma_{cm}$ seems to provide a good approximation of the lower bound of the lab rock long-term strength in a confinement range of 0–20 MPa. Near the underground excavation wall where the confinement is low (0–5 MPa), the slope of the failure surface is not relevant and $\sigma_1 - \sigma_3 = B\,\sigma_{cm}$ or $\sigma_1 = B\,\sigma_{cm}$ is adequate for the prediction of the onset of failure.

3.2.2. SKB ZEDEX

The Zone of Excavation Disturbance EXperiment (ZEDEX) project undertaken at the Äspö Hard Rock Laboratory (HRL), by the Swedish nuclear waste management company, was aimed at understanding the mechanical behavior of the Excavation Disturbed Zone (EDZ) with respect to its origin, character, magnitude of property change, extent and its dependence on excavation method [18]. The test site was located at the same depth as the URL, i.e., at 420 m. A number of in situ measurements including seismic tomography, borehole radar, hydraulic testing and acoustic emission monitoring were carried out as an integrate part of the experiment.

Rock mechanical laboratory testing of cores from boreholes shows that the Äspö diorite has a uniaxial compressive strength of 195 MPa with a standard deviation of 31 MPa. The crack initiation strength σ_{ci}, at about 64 MPa, was determined from acoustic emission techniques in the laboratory. The long-term rock strength (σ_{cd}), corresponding to the volumetric strain reversal, is 146 MPa [18].

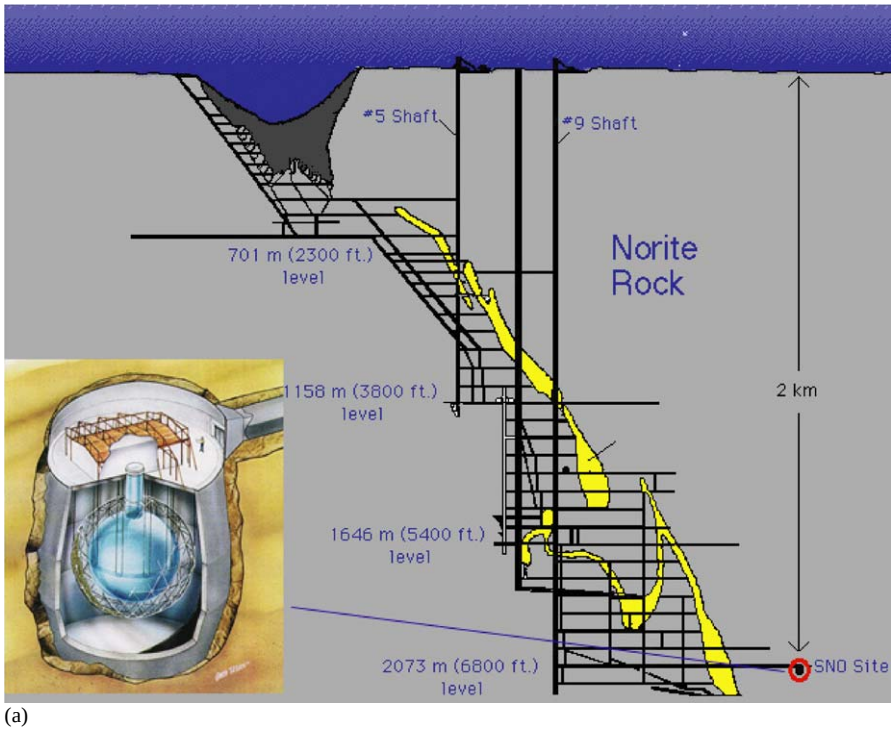
The rock mass at the HRL is moderately jointed. The rock mass quality in terms of *RMR* [54] were usually of Class II (*RMR* in the range of 61–80). Only small sections belong to Classes I and III (i.e., very good quality and fair quality). The mean *RMR* values obtained from different boreholes for the diorite range from 71 to 76 with a global mean of 72. The corresponding *Q* [55] values are in the range of 19–29 with a global mean of 24. The average *GSI* value estimated from the mean *RMR* or *Q* value is 73. The uniaxial compressive strength (σ_{cm}) of the rock mass obtained from the *GSI* system is 45.3 MPa with $m_i = 28$ for diorite.

AE monitoring was performed when TBM excavation was stopped or after each blast round in the blast excavated section. The spatial distribution of AE events for the two excavation methods was compared and it was found that the event rate was very high in the drill and blast section (about 10 times higher compared to the TBM excavation section). The stress at AE event locations was calculated using EXAMINE^{3D}, a boundary element program developed at Rocscience Inc. The average in situ crack (AE) initiation stress was found to be $\sigma_1 - \sigma_3 \approx 25$ MPa, or $\sigma_1 - \sigma_3 \approx 0.55\sigma_{cm}$ ($A = 0.55$). Note that if the non-generalized crack initiation criterion (Eq. (1)) is used, the stress level is given as $\sigma_1 - \sigma_3 \approx 0.13\sigma_c$. This results puzzled the SKB researchers because they had expected a crack initiation stress level comparable to that observed at the URL at about $\sigma_1 - \sigma_3 \approx 0.33\sigma_c$ or $\sigma_1 - \sigma_3 \approx 64$ MPa [18].

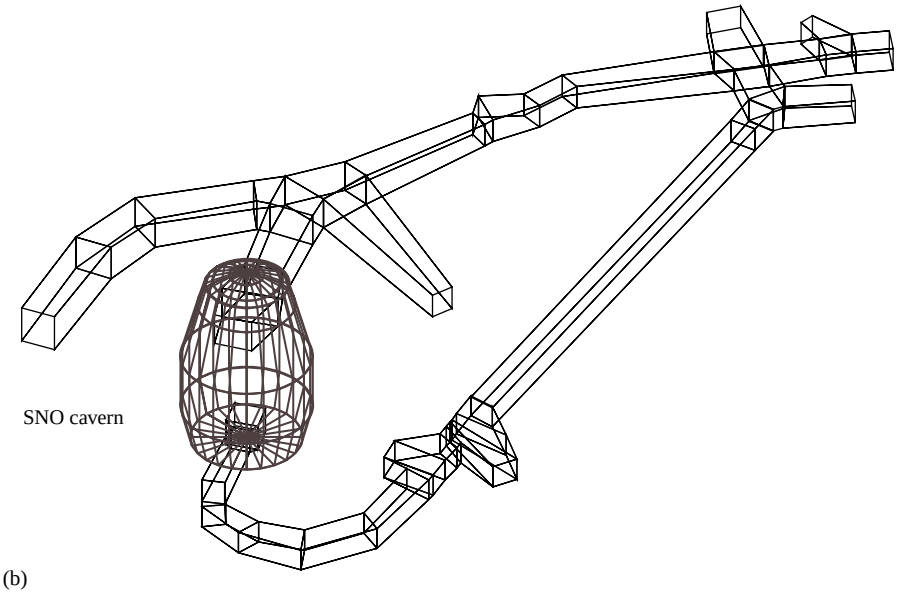
The crack initiation stress level of intact rocks is 64 MPa, which is much higher than the crack (AE) initiation stress level of the rock mass. It is seen that the initiation of new cracks in the intact rock due to increased differential stress was not a major cause of AE in the jointed rock mass around underground excavations at the HRL. Many events are within the zone of low confining stress, indicating that spalling initiation and failure may be playing a role in rock mass damage. Source mechanism analysis of AE events suggests that the great majority of events could be fit to shear-slip mechanism [18]. Whereas at the URL the predominant mechanism for the massive granite was tensile cracking [56,57].

3.2.3. SNO

The Sudbury Neutrino Observatory (SNO) cavern, which has a barrel shape with a maximum diameter of 23 m at mid-height and maximum height of 30.8 m, was placed at the 2070 m depth at INCO's Creighton Mine (Fig. 8), located in Sudbury, Ontario, Canada. The purpose of the facility is to detect neutrinos, which are elementary particles emitted from the Sun, in order to know what is happening in the Sun's core. The overburden rocks provide an excellent filter to eliminate other particles but will let the neutrinos to penetrate



(a)



(b)

Fig. 8. Creighton Mine and SNO: (a) plane view of the Creighton Mine and artistic drawing of SNO (http://www.sno.phy.queensu.ca/); (b) perspective view of SNO cavern and access ramps.

through. The excavation of the cavern was started in 1990 and completed in 1993. First neutrinos were detected in May 1999 and production data taking begins in November 1999.

The host rock surrounding the cavern is a fine grained norite which contains three major sub-vertical joint sets and one sub-horizontal joint set. Joints are tight and dry. *RMR* rating ranges from a lower bound of 61 to an upper bound of 81 [58]. Extensive monitoring, which include internal and external extensometers, CSIRO stress cells, borehole camera, and microseismic system, were used to record the rock mass response during cavern excavation. The extent of the damaged zone was identified from the borehole camera observation, extensometer and stress cell monitoring data. The principal stresses at the monitoring locations were obtained from an elastic analysis by MAP3D [59]. Their analysis results showed that the crack damage threshold

was defined as $\sigma_1 - \sigma_3 \approx 70-80$ MPa, or $\sigma_1 - \sigma_3 \approx (0.31–0.35)\sigma_c$, where $\sigma_c = 225$ MPa.

It should be kept in mind that the rock mass at the SNO cavern site contains joints. The rock mass quality assessment is made from exposed rock masses at drifts which have been disturbed by blasting. The upper bound of *RMR* is used to represent the undisturbed rock mass. When a *GSI* value of 81 is applied, along with $\sigma_c = 225$ MPa and $m_i = 33$ for norite, the equivalent rock mass strength σ_{cm} is approximately 80 MPa. If we replace σ_c by σ_{cm} in the equations above, then we obtain a crack damage threshold of $\sigma_1 - \sigma_3 \cong (0.88–1.0)\ \sigma_{cm}$ ($B = 0.88–1.0$) for the jointed rock masses.

For this case, the crack (AE) initiation threshold cannot be calibrated because detailed microseismic monitoring data are not available. If microseismic data were available, we would anticipate a crack initiation threshold of $\sigma_1 - \sigma_3 \approx (0.4–0.5)\ \sigma_{cm}$ or $\sigma_1 - \sigma_3 \approx 32$ to 40 MPa for $\sigma_{cm} = 80$ MPa.

3.2.4. Creighton mine

Creighton Mine, located in Sudbury, Ontario, Canada, is the largest and deepest operating nickel mine in North America (Fig. 8). For deep mining level at Creighton Mine, the in situ crack damage threshold was calibrated [17] using modeling simulation and field observation data. Borehole camera and extensometer information were used in combination with drilling logs to delineate the extent of the damaged and fractured zone. MAP3D was used to simulate the stress field and the stress states at all observation points were calculated and plotted [22]. The damage stress level in the form of Eq. (1) was found to be applicable. The damage threshold was established at $\sigma_1 - (1.0–1.4)\ \sigma_3 \approx 90–100$ MPa or $\sigma_1 - \sigma_3 \approx (0.4–0.5)\ \sigma_c$ when normalized to an intact rock strength σ_c of 250 MPa. The upper limit of this range corresponds to locations where rock mass damage could be observed by visible observations in boreholes.

It should be kept in mind that the rock mass at the Creighton Mine site contains joints. The rock mass quality is very good with a *GSI*-value of 80–85, $\sigma_c = 250$ MPa and $m_i = 33$, resulting in an equivalent rock mass strength σ_{cm} of 86–112 MPa. If we replace σ_c by σ_{cm} in $\sigma_1 - \sigma_3 \approx (0.4–0.5)\ \sigma_c$, we obtain a damage threshold of $\sigma_1 - \sigma_3 \approx (0.8–1.0)\ \sigma_{cm}$ for the jointed rock masses.

Recently, a study is undergoing at MIRARCO to visualize and interpret the microseismic data collected at Creighton Mine. 3D stress analysis is performed to obtain the principal stresses at the event locations. A preliminary analysis indicates that the damage initiation threshold is around $\sigma_1 - \sigma_3 \approx 52$ MPa or $\sigma_1 - \sigma_3 \approx 0.53\sigma_{cm}$. Detailed study results will be made available in a separate publication.

3.2.5. Löetschberg tunnel

In Switzerland, two of the world's deepest railway tunnels are under construction through the Alps. The Lötschberg Base Tunnel, connecting Bern in Switzerland to Milan in Italy, is part of the Alp Transit concept. It is 34.6 long from Frutigen in the Kander valley to Raron in the Rhone valley. The Lötschberg tunnel construction started in October 2000 and the tunnel is scheduled to open in July 2006.

The southern lots of the tunnel are being excavated by TBM. Due to the high overburden (3000 m or more), the vertical stress is high [60]. Localized spalling from the sidewalls and minor rockbursting are observed. For the massive granite along the tunnel, it is documented [60] that $\sigma_c = 200$ MPa, $m_i = 20$, and $GSI = 88–90$, and the corresponding rock mass strength σ_{cm} is 103–115 MPa, with an average of about 110 MPa. From km 3.0–4.5 of the Steg entry, the calculated maximum wall stress varies from 88 to 105 MPa. Isolated spalling failure was observed in the areas where the overburden is greater than 2700 m. Due to the natural strength variation, spalling is only encountered at places where the maximum wall stress exceeds the wall strength. This is similar to the case at the URL Mine-by tunnel, but at the URL the rock is homogenous so that continuous notch failure along the tunnel was observed. The crack (AE) damage threshold at the Löetschberg tunnel is found to be on average $\sigma_1 - \sigma_3 \approx (0.8 \text{ to } 0.95)\sigma_{cm}$, resulting in an B value between 0.8 and 0.95.

Because no AE/MS monitoring data is available, the crack initiation threshold cannot be precisely determined. However, by analogue to the URL case it is anticipated that $\sigma_1 - \sigma_3 \approx 66$ MPa or $\sigma_1 - \sigma_3 \approx 0.6\sigma_{cm}$ defines the crack (AE) initiation at the Löetschberg tunnel. As at the URL, the depth of failure corresponds with the depth of the $\sigma_1 - \sigma_3 \approx 66$ MPa or $\sigma_1 - \sigma_3 \approx 0.6\sigma_{cm}$ contour. The two stress thresholds at the Löetschberg tunnel and URL are comparable.

3.2.6. Kannagawa powerhouse cavern

The Kannagawa pumped hydropower project [7,61] in Gumma Prefecture in Japan is now under construction with a maximum output of 2700 MW. The powerhouse cavern at 500 m depth has a width of 33 m, a height of 52 m, and a length of 216 m. The cavern excavation was started in 1998 and the last bench was completed in 2000.

An extensive AE study program was executed during the construction of the underground powerhouse, which included AE monitoring of triaxial tests on samples taken from the construction site, and in situ AE monitoring during cavern excavation. Before cavern excavation started, a total of 120 AE sensors were installed along 10 monitoring lines, equally divided in two cross-sections (D and H). The layout of AE sensors in Section H is presented in Fig. 9. All sensors were

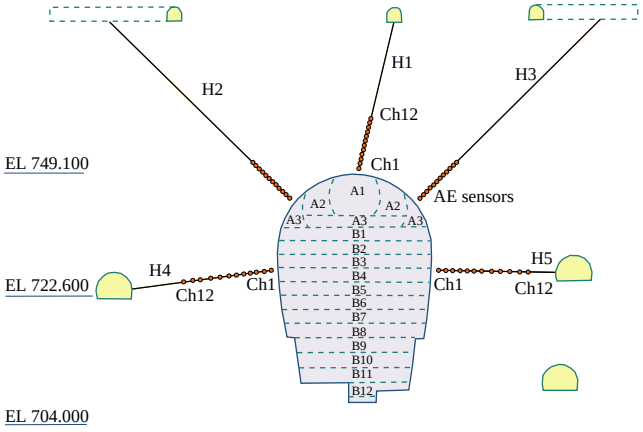


Fig. 9. Layout of AE monitoring sensors in H section. Sensors are numbered sequentially from shallow to deep locations.

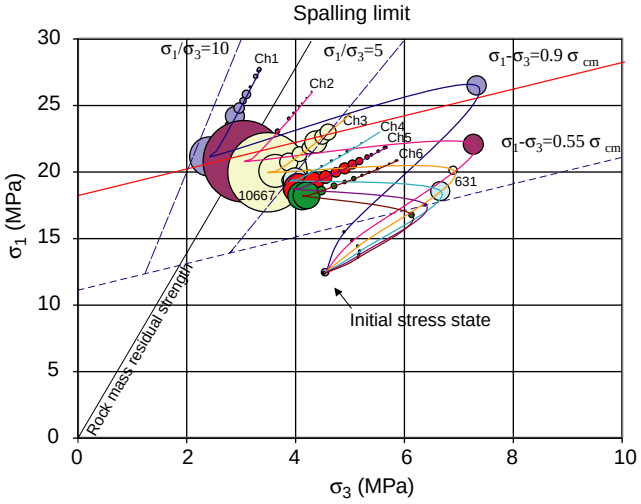


Fig. 10. Crack (AE) initiation and damage at line H3 for Ch1 to 6 (1.5–6.5 m).

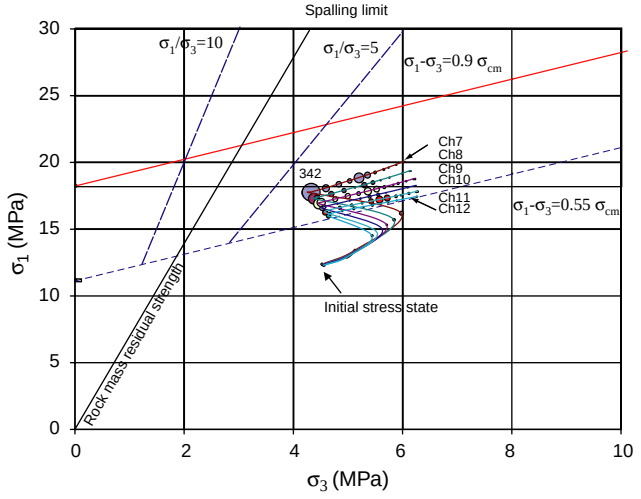


Fig. 11. Crack (AE) initiation and damage at line H3 for Ch 7 to 12 (7.5–12.5 m).

20 mm in diameter, 26 mm in height with a resonance frequency of 70 kHz. The sensors located in the rock mass on the sidewalls (H4, H5) recorded event data only during bench excavations. 2 h of continuous data acquisition was conducted after each bench block blast. Data were processed automatically to eliminate noise. AE events were then processed using FFT and recorded on hard drives for later interpretation and analysis.

The rock mass at the Kannagawa site is characterized using the *GSI* system [50]. *GSI* values for conglomerate, sandstone, and mudstone range from 54 to 74 and the equivalent rock mass uniaxial compressive strengths range from 5 to 29 MPa.

FEM elastic stress analysis was performed to calculate the principal stresses at each sensor location and the stress paths are plotted along with the AE event counts to identify AE (crack) initiation and propagation (crack damage) thresholds. AE initiation threshold was identified from the plot when systematic AE events started to appear along the stress path. The identification of the AE damage threshold was assisted by the assessment of the AE center frequency changes at each sensor location. The center frequency is the frequency at the spectrum center after a FFT transformation of the waveform from the time domain to the frequency domain. Two examples are presented in Figs. 10 and 11. The results for other measurement lines show consistent result and the details will be published in the future. The area of the circles in these figures indicates the number of AE events generated at the sensor location during bench excavation. The spalling limit [16], defined by a range of $\sigma_1/\sigma_3 = 5$–10, and the rock mass residual shear strength, defined by $\tau = 0 + \sigma \tan 49^\circ$ from in situ block shear test, are also shown in these figures.

Line H3 is situated in a zone with a rock mass strength of $\sigma_{cm} = 20.2$ MPa. When bench A2 is excavated, the stress paths at sensor locations Ch1 to Ch4 cross the crack initiation threshold $\sigma_1 - \sigma_3 = 0.55\sigma_{cm}$ (Fig. 10). Ch5 and Ch6 cross the crack initiation threshold when bench A3 is excavated, and Ch7 cross the line when bench B1 is excavated (Fig. 11). The AE (crack) damage threshold defined by $\sigma_1 - \sigma_3 = 0.9\sigma_{cm}$ indicates that rock mass behavior at Ch1 to Ch3 has exceeded the damage threshold. This observation is in agreement with the monitoring results from extensometers showing that the displacement at these three sensor location deviate from the elastic response. It follows that even for the moderately jointed rock mass with a compression strength between 20 and 29 MPa, the generalized crack (AE) initiation and crack damage criteria apply.

The crack damage zone defined from the generalized crack damage threshold is in good agreement with the

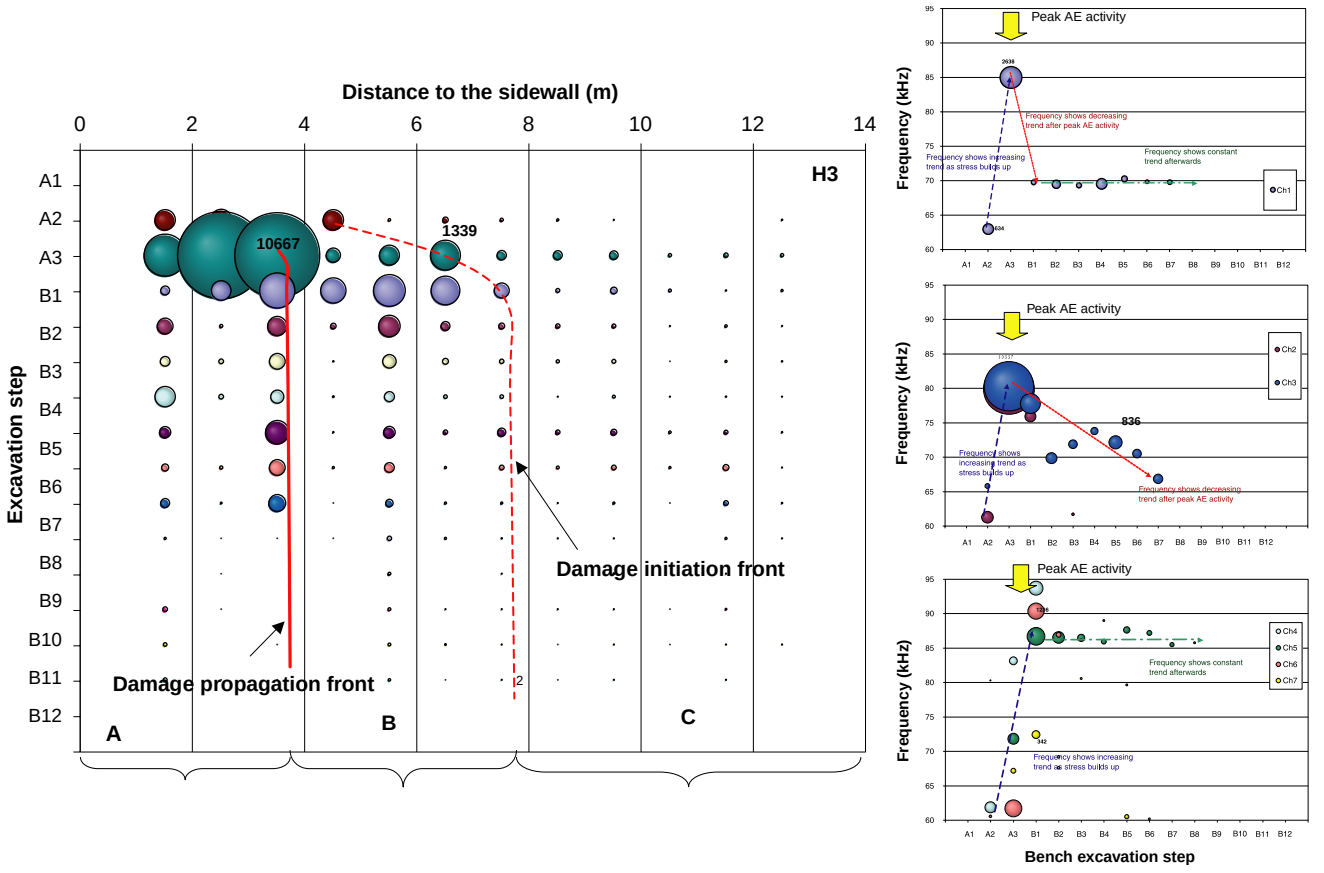


Fig. 12. Evolution of AE events at measurement line H3 during bench excavation and the correlation between AE counts and AE center frequencies.

zone determined from the AE counts and AE center frequency analysis shown in Fig. 12. Fig. 12 shows the evolution of AE events along measurement line H3 during bench excavation. The damage propagation front is determined by examining AE count and AE center frequency changes at each sensor location as the excavation advances. In general, AE center frequency reaches peak near the peak strength, which is associated with the AE event (rate) peak in most cases. After event peaks, there is a tendency that the center frequency decreases as the event number decreases. At line H3, the AE initiation depth reaches 4.5 m when A2 is excavated. This zone extends to 6.5 m when A3 is excavated, and reaches a maximum of 7.5 m when B1 is excavated. When A2 is excavated, the event peak happens at Ch2 and Ch3. The event peaks for other excavation steps are mostly at Ch3 sensor location, indicating that Ch3 location is the front of stress concentration. In Ch1 to Ch3, the frequency drop trend after AE event peaks is evident (Fig. 12). In other channels, the center frequencies show an increasing or constant trend throughout the excavation process, indicating that stresses in these areas and beyond are increasing or at least are maintained. Hence, it is concluded that the damage zone depth is about 3.5 m at line H3.

Furthermore, the data from this monitoring section supports the concept of spalling-type failure mechanism when the spalling limit [16] is exceeded. As soon as the stress path enters the spalling limit, the rate of AE events increases rapidly and the AE event number for each bench excavation (represented by the area of circle) drastically increase.

4. Conclusions

Generalized crack initiation and crack damage thresholds are proposed in the present study. By replacing the intact rock uniaxial compressive strength σ_{cd} by the uniaxial compressive strength σ_{cm} of the jointed rock mass, the crack initiation ($\sigma_1 - \sigma_3 = A\sigma_{cm}$) and crack damage ($\sigma_1 - \sigma_3 = B\sigma_{cm}$) thresholds take a general form that is found to be applicable to massive, moderately jointed and jointed rock masses of fair or better quality. The rock mass strengths determined from the *GSI* system can be used to establish the material constants A and B. For massive to moderately jointed rock masses, A and B are found to be in the range of 0.4 to 0.5, 0.8 to 0.9, respectively, and for moderately to highly jointed rock masses, A and B are in the range of 0.5 to 0.6, 0.9 to

1.0, respectively. This approach is not applicable to plastically yielding rocks that do not fail by crack or fracture damage accumulation.

The crack initiation and crack damage surfaces are linear in the σ_1 and σ_3 space. From laboratory tests on intact rocks, it is seen that these criteria are true in low confinement conditions (typically <5 MPa). The case histories of crack (AE/MS) initiation and damage investigation presented in this article confirm that the simple form of linear thresholds provide a practical means to identify when instabilities are to be anticipated in underground construction.

To assess the rock mass crack initiation and crack damage stress levels, it is important to obtain a reliable assessment of the rock mass strength. This can be achieved through applicable field rock mass tests or, alternatively, the application of the *GSI* system.

These generalized criteria are applicable to low confinement conditions, i.e., rock failure or damage of the rock mass near an underground excavation. When combined with linear elastic stress analysis, these crack initiation and crack damage thresholds assist in assessing the rock mass integrity.

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